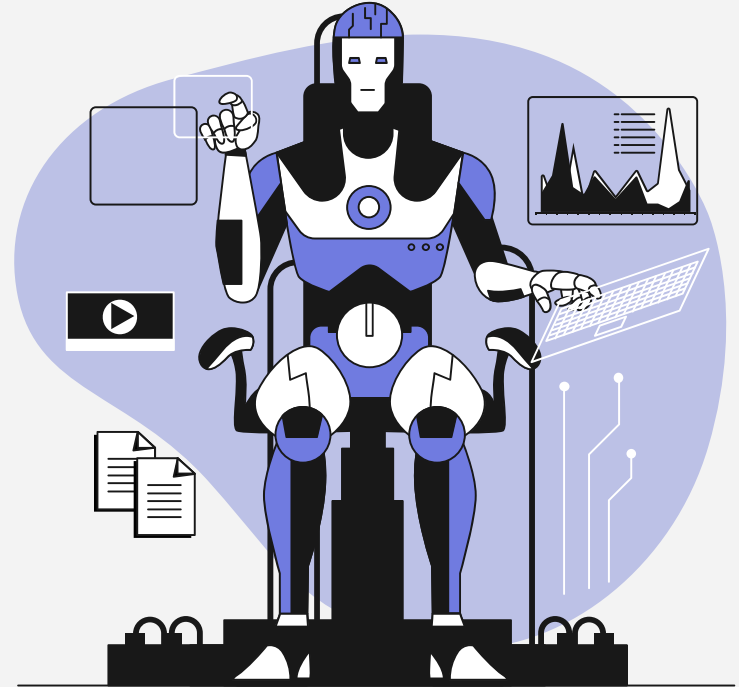


Decision Tree

Machine Learning - HAM Arc





Agenda

01

Introduction

What is tree node, leaf,...

02

Purity & Entropy

Purity, Entropy function,
graph, uses

03

Information Gain

What is information gain
and its uses

04
Tree

Ensemble

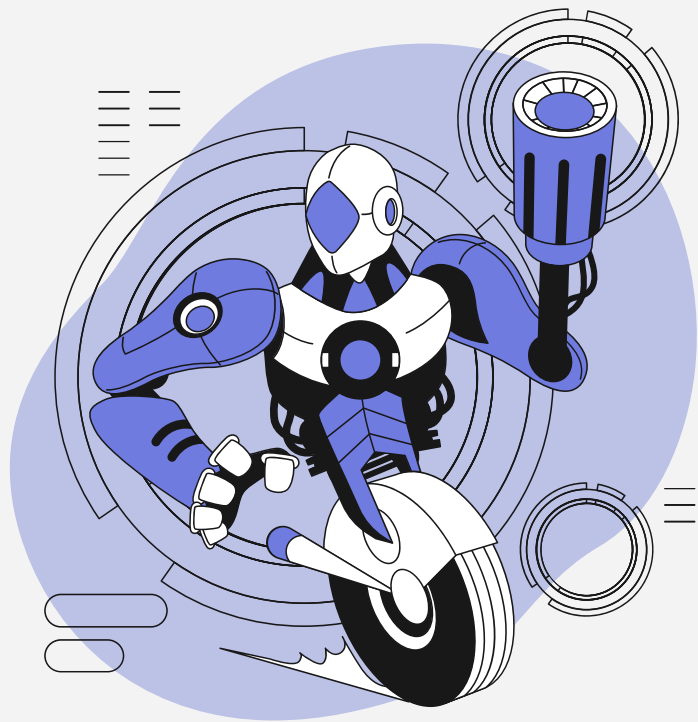
What is tree ensemble

05

Generating Tree Samples

Bagged decision trees,
random forest, XGBoost

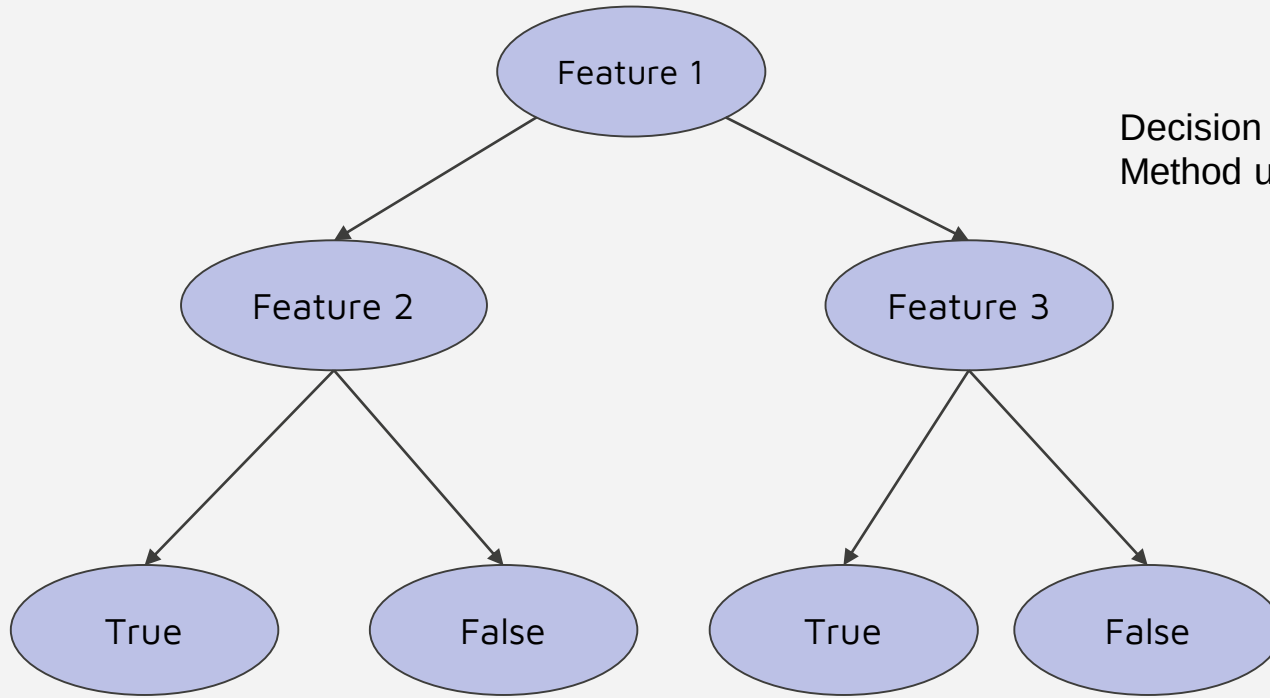




01

Introduction to Decision Tree





Decision tree is a supervised learning algo/
Method used to train a model

Feature 1 is the root node
of the decision tree

Feature 1,2,3 are the
decision nodes of the tree

Last 4 nodes are the leaf
nodes



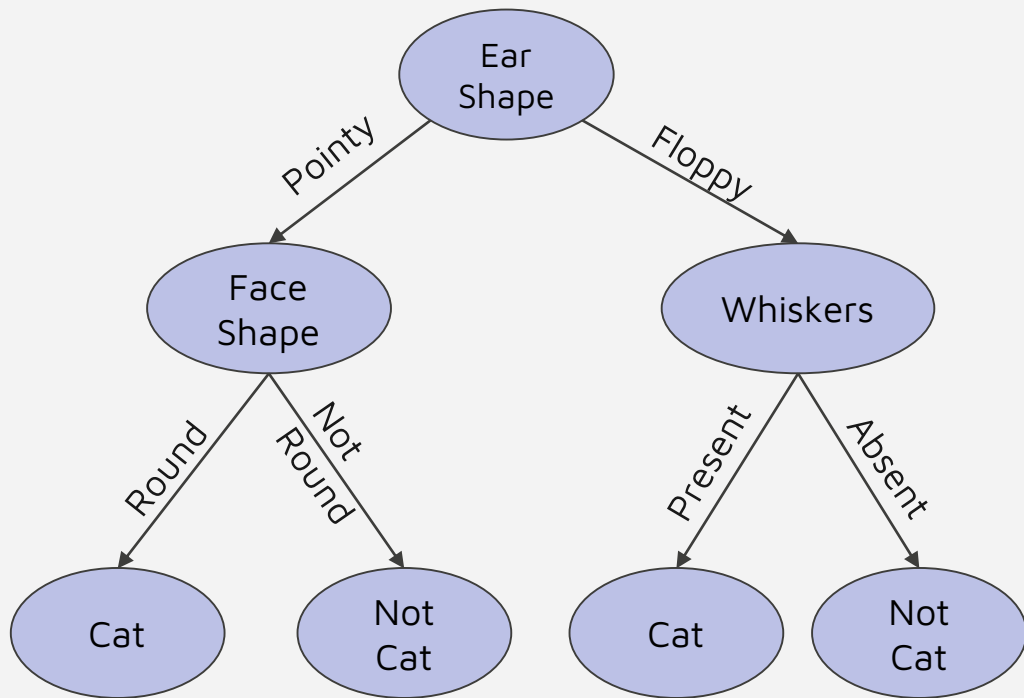
Example

Ear shape (x1)	Face shape (x2)	Whiskers (x3)	Cat
Pointy	Round	Present	Yes
Floppy	Not Round	Present	Yes
Floppy	Round	Present	No
Floppy	Round	Absent	No



X

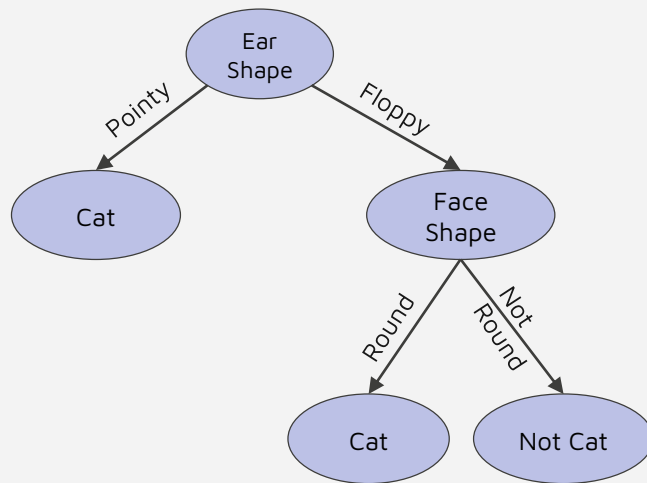
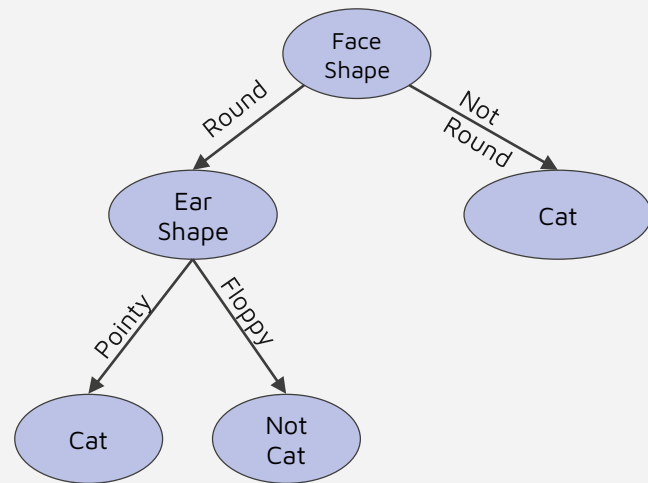
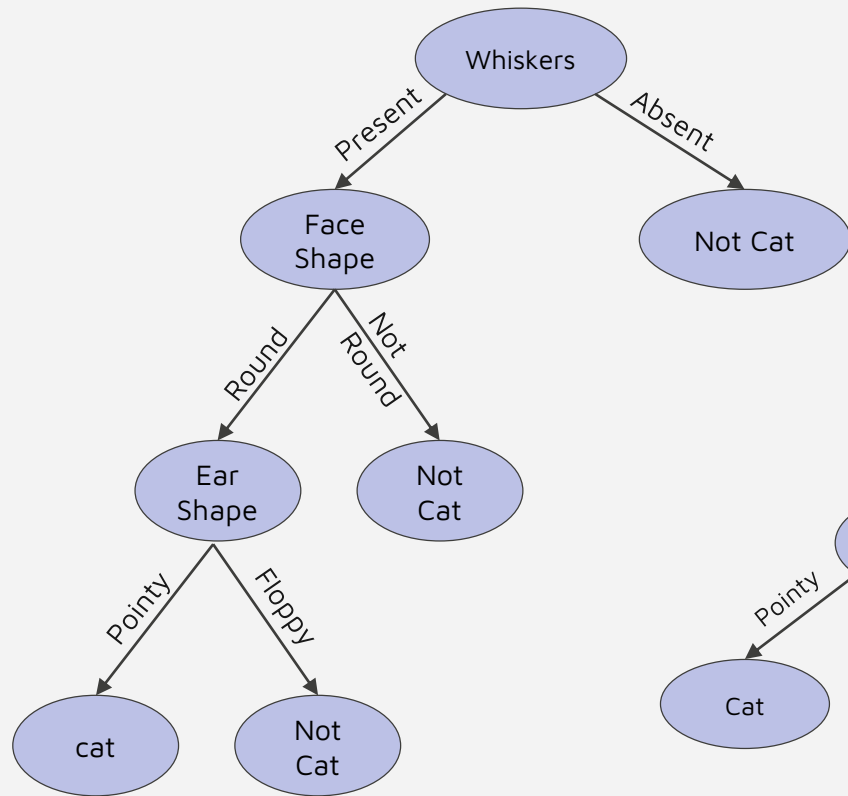
Y



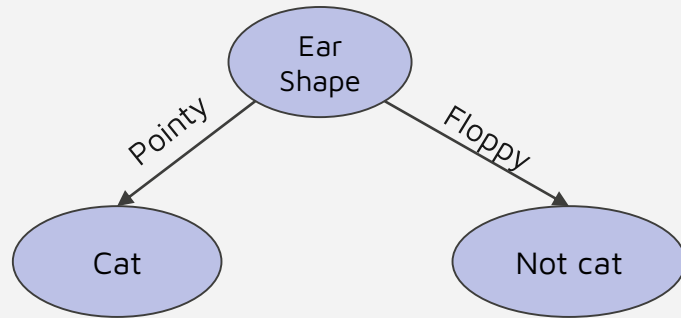
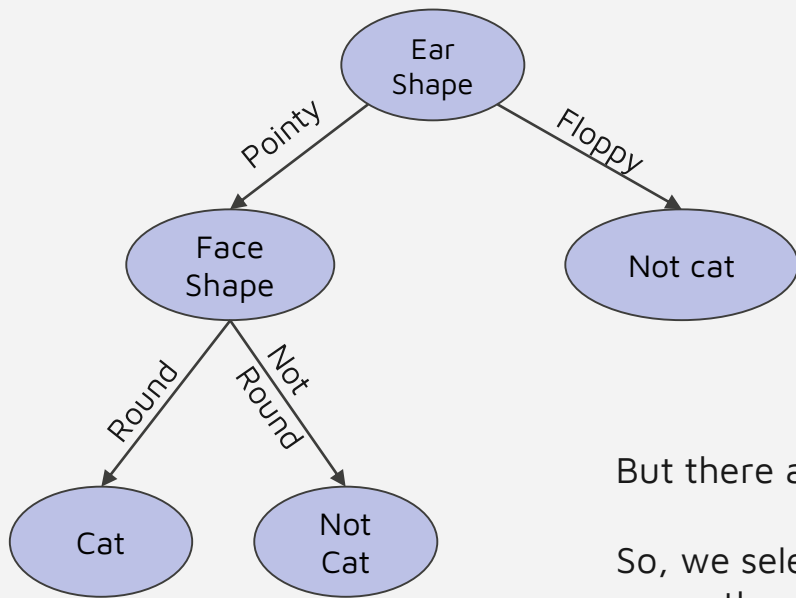
Is this the only possible tree?

NO!

Ear shape (x1)	Face shape (x2)	Whiskers (x3)	Cat
Pointy	Round	Present	Yes
Floppy	Not Round	Present	Yes
Floppy	Round	Present	No
Floppy	Round	Absent	No



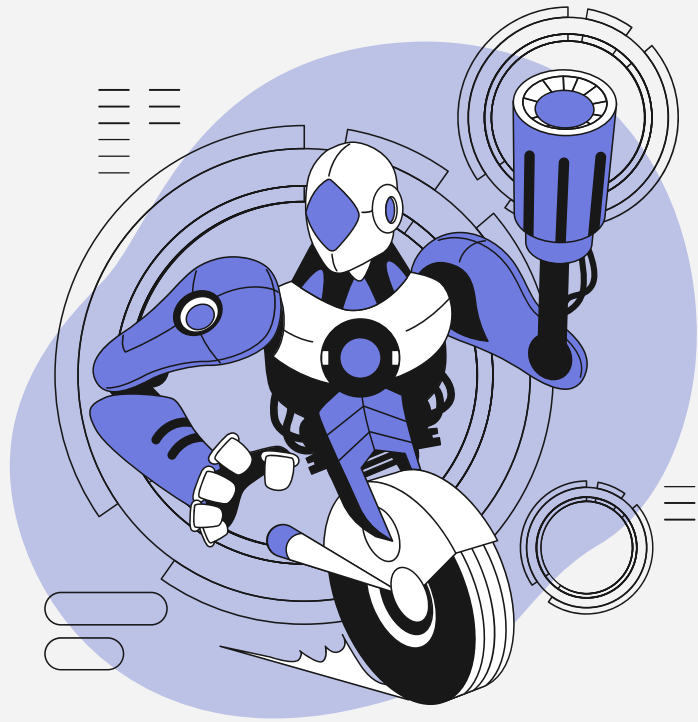
And many more...
Set of these trees are called
a **FOREST/TREE ENSEMBLE**



But there are few trees which doesn't correctly follow the training set

So, we select features at each node such that the resulting decision tree correctly classifies all the training data. A tree that perfectly classifies all training examples is called a perfect (or pure) decision tree.

To do that we need to know about **PURITY, ENTROPY and INFORMATION GAIN...**



02

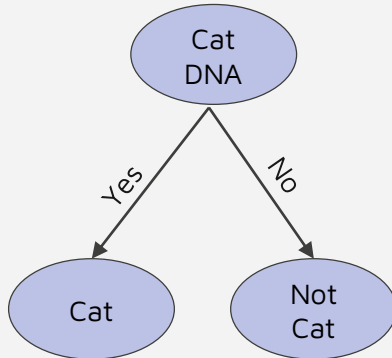
Purity & Entropy

What is purity?

Purity tells us how “mixed” or “pure” the node is

Formula:

$$Purity = \frac{\text{number of samples in the majority class}}{\text{total samples}}$$



Eg:

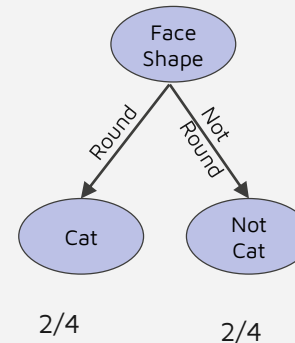
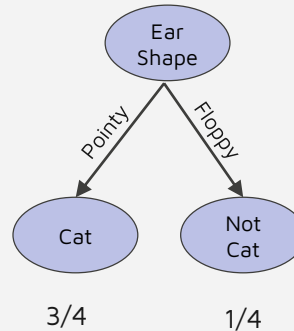
Pass	Fail	Purity
10	0	10/10=1 ie 100%
4	6	4/10=0.4 ie 40%

100% purity, that means all examples reaching that node belong to the cat, so the decision at that node is certain according to the training data

Benefits of Purity

- 1) Helps to choose root node
 - The node with maximum purity is chosen as the root node
- 1) Helps to decide when to stop the algorithm
 - When the node obtains 100% purity
 - When splitting reaches the maximum depth (the factor is decided by the developer)

Ear shape (x1)	Face shape (x2)	Cat
Pointy	Round	Yes
Pointy	Round	Yes
Floppy	Not Round	No
Pointy	Not Round	No

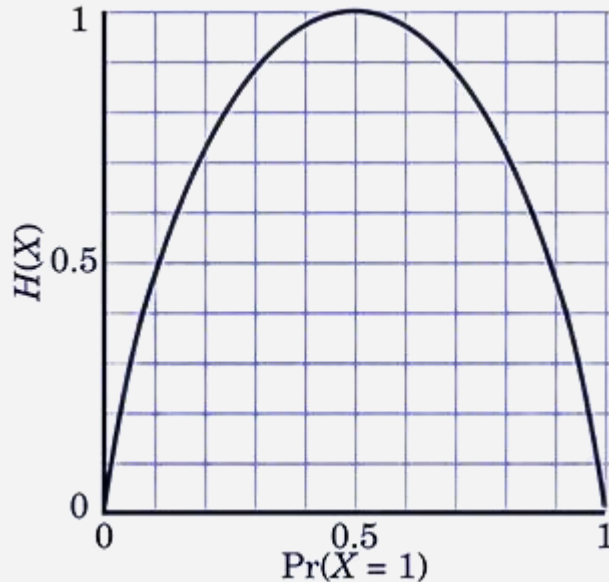


NOTE:

Keep the tree small to avoid overfitting

Entropy

Entropy is a metric used in machine learning (specifically in decision trees) to measure the impurity, disorder, or randomness in a dataset node



1. Maximum Impurity ($H = 1$): If a node has a perfectly even 50/50 split of classes (e.g., 50% Cats and 50% Not Cats), entropy is at its maximum of 1. This represents maximum uncertainty/disorder.
1. Maximum Purity ($H = 0$): If a node contains only a single class (e.g., 100% Cats or 0% Cats ie 100% Not Cats), entropy drops to 0. This represents complete certainty.

Therefore,
Less the entropy = More purity & More the entropy = Less purity



Entropy Function

$$H(p_1) = -p_1 \log_2(p_1) - p_0 \log_2(p_0)$$

$$p_0 = 1 - p_1$$

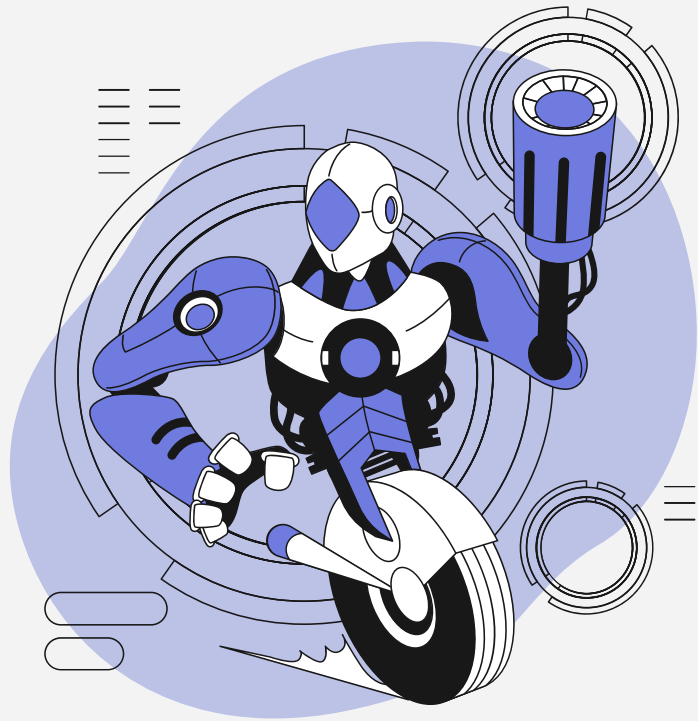
so,

$$H(p_1) = -p_1 \log_2(p_1) - (1 - p_1) \log_2(1 - p_1)$$

NOTE:

Here $\log(0)$ is undefined, so just for convention we take $\log(0)=0$



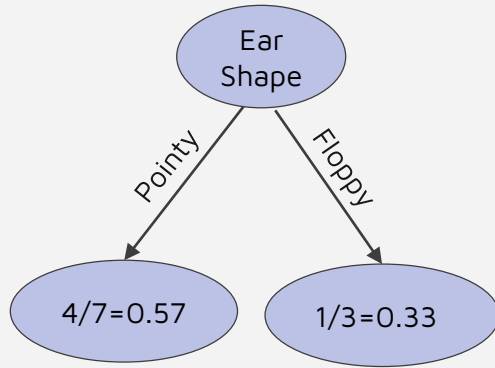


03

Information Gain

Information Gain

$$\text{Information Gain} = H(p_1^{\text{root}}) - [w^{\text{left}} H(p_1^{\text{left}}) + w^{\text{right}} H(p_1^{\text{right}})]$$



$$\begin{aligned} &= H(0.5) - [(0.57)H(0.57) + (0.33)H(0.33)] \\ &= 1 - [0.57(0.9858) + 0.33(0.9149)] \\ &= 1 - 0.9646 \\ &= 0.0354 \end{aligned}$$

Tells us about how much a particular split improves the purity

Accordingly you will calculate Information Gain for all root nodes and the maximum will be the root node

Benefits of Information Gain

- 1) Helps to choose root node
 - The node with maximum information gain is chosen as the root node
- 1) Helps to decide when to stop the algorithm
 - Information Gain is less than the threshold or number of the examples is less than threshold

When else to stop the algorithm?

- When a node with 100% purity ie 0% entropy is achieved
- When one more feature added will exceed the predefined depth by developer

One Hot Encoding

Converting the feature values into zeros and ones, done because when a machine-learning algorithm needs numerical input and the categories have no natural numerical order

Ear shape (x1)	Face shape (x2)	Cat
Pointy	Round	Yes
Pointy	Round	Yes
Floppy	Not Round	No
Pointy	Not Round	No

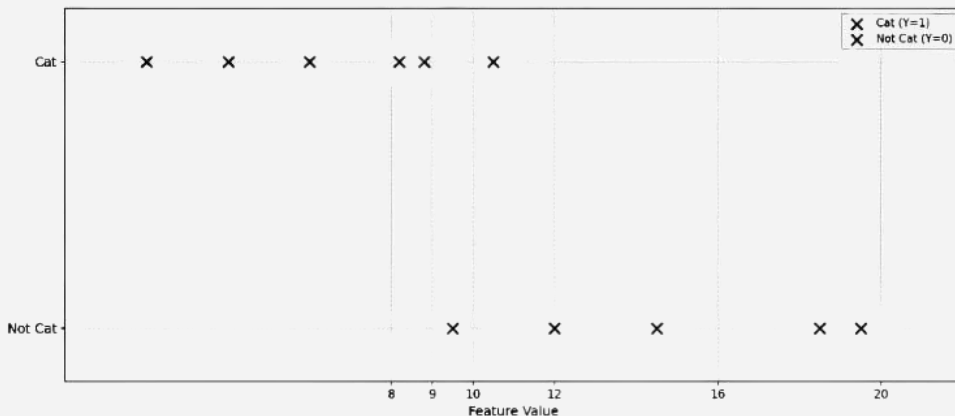


Ear shape (x1)	Face shape (x2)	Cat
1	1	1
1	1	1
0	0	0
1	0	0

Splitting on a continuous variable

Weights of cats
8.8
9.2
15.8
11.4
.....

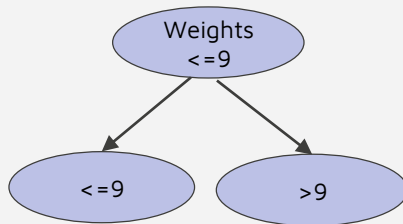
Plot the values in a graph, we may get a graph like this

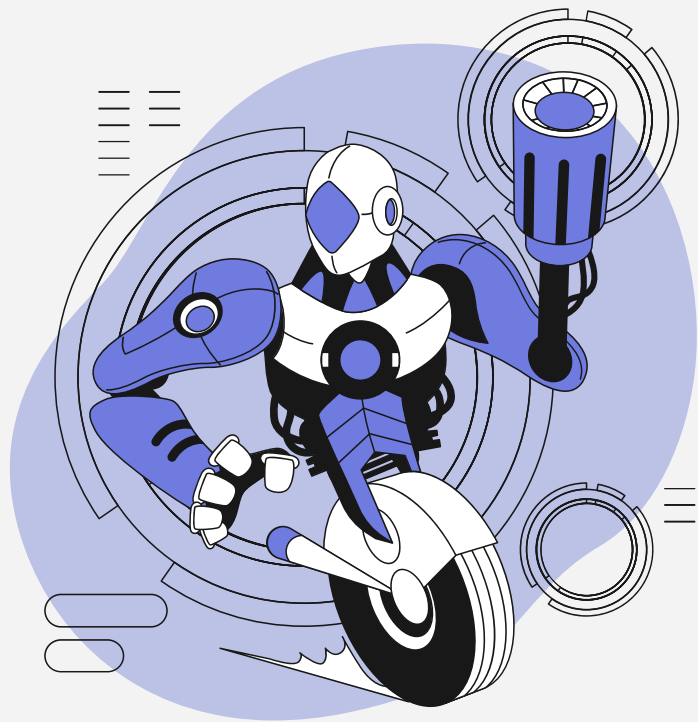


Guess the value such that it covers both left and right equivalently, calculate the information gain, the maximum value is the root

Here,
information gain (IG) at 5=0.31
 $IG(9)=0.61$, $IG(12)=0.27$

Information Gain at 9=0.61
which is highest, therefore the
tree is defined as follows



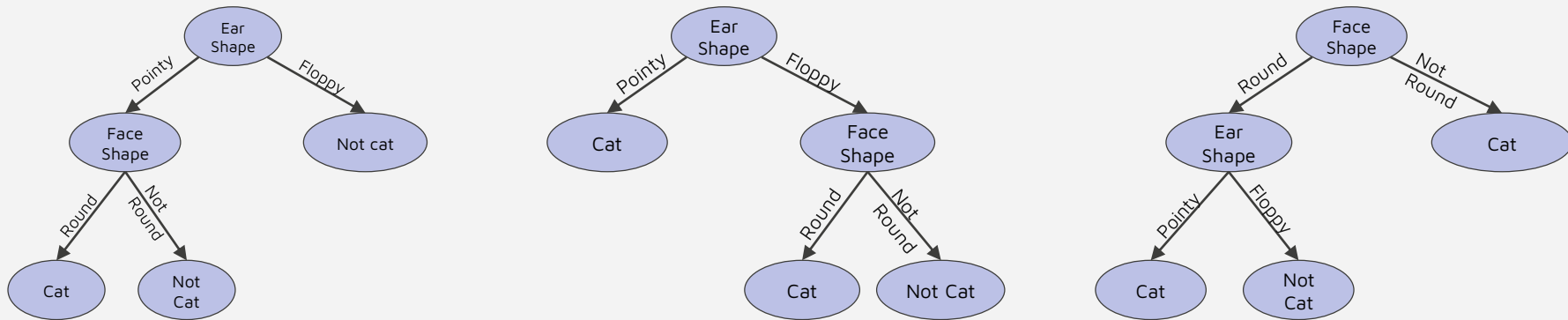


04 Tree Ensemble

Tree Ensemble

Just one extra dataset can change the structure of the tree because the information gain of the entire tree will change, in worst case the tree's root can change too, so it's better to have many decision trees, they are called tree ensemble,

So instead of relying on one potentially unstable tree, we can build many different decision trees and combine their predictions.

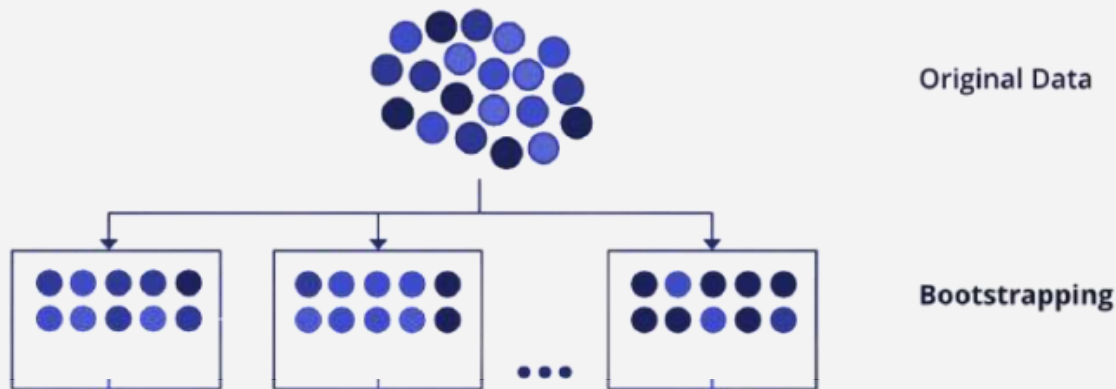


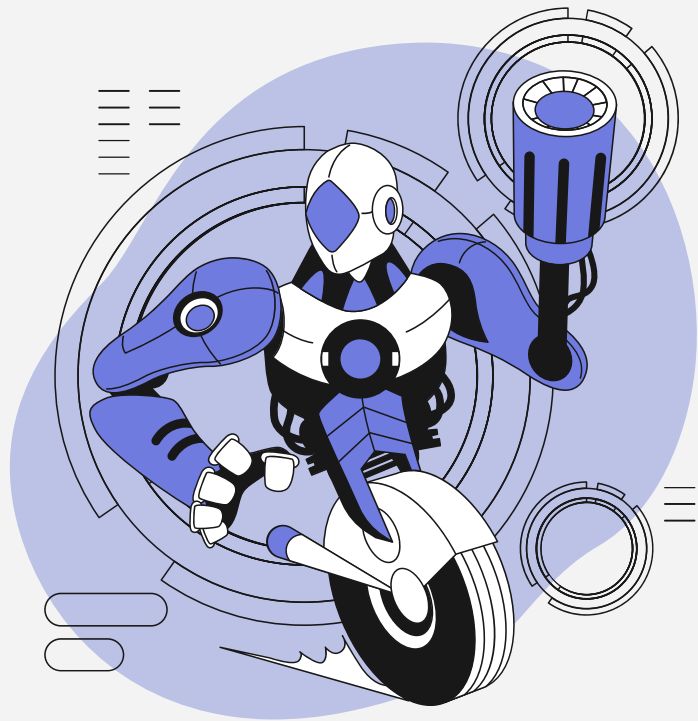
Tree 1 and 3 predicts True, but Tree 2 predicts False, therefore, as per majority the prediction is True

Sampling using replication

This method is also known as **Bootstrapping**

Basically you imagine that you put all your training dataset in a bag, and you pick one dataset at a time, write it down & keep it in the bag again. Repeat the process. Take as much as readings you want & create a decision tree as we have did before. Now create more trees & by the end we will have a Tree ensemble





05

Generating Tree Samples

Bagged Decision Trees & Random Forest

Given a training set of size 'm',

b = 1 to B times the process is repeated.

- Using sampling with replacement method, various training sets of size 'm' can be created, using that create decision trees.

This will be done for 'B' times.

B = 64-128 (better to be < 100)

Trees created by this method is called Bagged decision trees.

At some training sets, the root node might be the same because one feature dominates, so we make a subset of all features whose size is less than m (say $k < m$, usually $k = \sqrt{m}$), The algorithm is forced to pick the best split only from this k-feature subset. This prevents trees from constantly picking the same root node and creates diverse trees. This algorithm is called a **Random Forest**.



XQBoost



XGBoost/Extreme gradient boosting/Boosted tree intuition, is the same as before, but the only difference is that, at places where there is a problem like it's not working / efficiency is less, it better focus on that first then look on other.

For $b = 1$ to B :

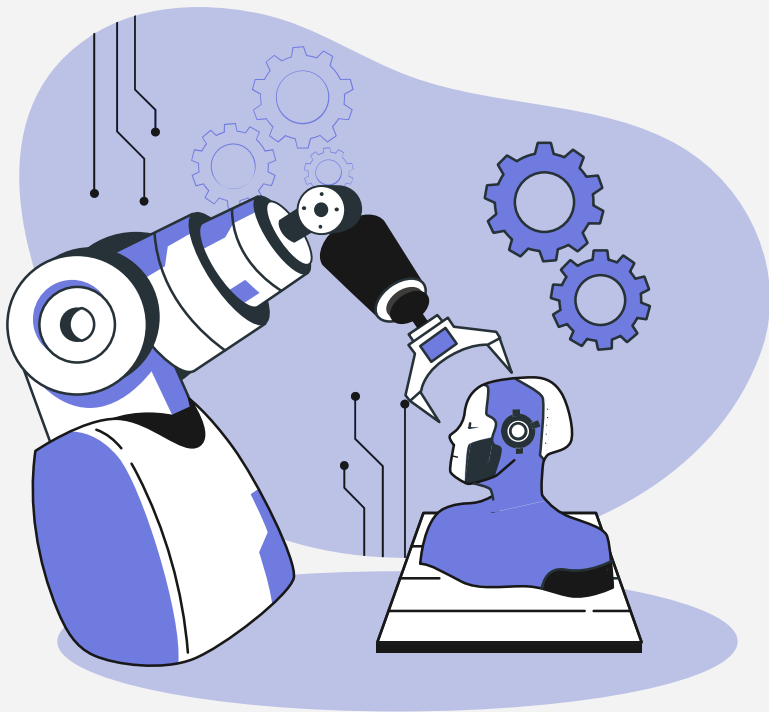
- Use sampling with replacement to create new training set of size ' m '.
- But instead of picking from all examples with equal ($1/m$) probability, make it more likely to pick misclassified examples from previously trained trees (more imp given for them).
- Now train a decision tree on new dataset.

So In this algorithm, each new tree sequentially focuses on the mistakes made by previous trees, iteratively refining the model to minimize total prediction error





Why XQBoost?



- Open source implementation of boosted trees.
- Fast efficient implementation.
- Good choice of default splitting criteria & criteria for when to stop splitting.
- Built in regularization to prevent overfitting.
- Highly competitive algorithm.



Decision Tree vs Neural Networks



Decision Tree

Works on only structured datas,
fast, Not very expensive



Neural Networks

Works with structured and
unstructured data ie videos,
audios, and all, slow, very
expensive

THE END